

Iterative Re-Linearisation for a Naturally-Binary QUBO Formulation of Low-Thrust Cislunar Trajectory Scheduling

Project status and complete Earth–Moon mission demonstration

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Outline

- 1 Context and motivation
- 2 The mission
- 3 Problem and method
- 4 Iterative re-linearisation
- 5 Phase 2: cislunar correction
- 6 Phase 4: station-keeping
- 7 Phase 3: sliding-window lunar capture
- 8 End-to-end mission
- 9 Sensitivity, comparison, future work
- 10 Conclusion

Context: cislunar logistics and low-thrust trajectories

Renewed interest in cislunar missions: lunar gateway, L1/L2 relay constellations, lunar surface logistics.

Demand for trajectory tools that scale to many revolutions and tight timing.

Direct collocation NLP and indirect Pontryagin methods are mature but rely on *continuous-variable* nonlinear programming.

Quantum annealing (QA) and the QUBO formalism have recently been applied to interplanetary trajectory optimisation (**De Grossi, Carbone et al., 2025** — *Astrodynamics* 9:195–215).

Carbone et al. baseline

Pseudospectral interpolation of state and control over the time interval, encode real-valued coefficients as binary digits, submit the resulting QUBO to D-Wave hybrid solvers. Earth–Mars results demonstrate QA reaching competitive solutions.

Two ways to put low-thrust on a QUBO solver

Carbone et al. (continuous-coefficient)

- Pseudospectral coefficients are real-valued.
- Encode each as p bits.
- Quantisation error $\sim O(2^{-p})$ per coefficient.
- Natural for arbitrary-shape thrust profiles.
- Best when inner least-squares is non-trivial.

This work (naturally-binary)

- Decisions are inherently binary: $q_i \in \{0, 1\}$ = engine on/off at step i .
- No quantisation error: variables are not approximations of a continuum.
- Direct fit to pulse-driven thrusters.
- N qubits for N time slots — linear, not quadratic.

Complementary niches

The two formulations target different aspects of low-thrust QA. This talk presents the second.

One spacecraft, one thruster, four phases

Spacecraft: ~ 100 kg microsatellite, single ion/Hall-class thruster, native throttling.

Real comparison: NEXT-C (25–235 mN), BepiColombo T6 (75–145 mN), DS4G (50–2500 mN).

Operating points used

Main power: 350 mN, $a \approx 3.5 \text{ mm s}^{-2}$.

Cruise power: 5.4 mN, $a \approx 0.054 \text{ mm s}^{-2}$.

Single physical system, two of its native operating points.

Phases

Phase 1	Earth ascent (GEO $\rightarrow$ parking $\rightarrow$ injection)	main, tangential
Phase 2	Cislunar correction (QUBO)	cruise, tangential
Phase 3	Lunar capture (sliding-window QUBO)	main, anti-tangential
Phase 4	Lunar-orbit station-keeping (QUBO)	cruise, anti-tangential

Planar CR3BP and the scheduling decision space

Equations of motion (synodic frame, Earth at $(-\mu, 0)$, Moon at $(1 - \mu, 0)$, $\mu \approx 0.01215$):

$$\ddot{x} - 2\dot{y} = \Omega_x(x, y) + u_x$$

$$\ddot{y} + 2\dot{x} = \Omega_y(x, y) + u_y$$

where $\Omega(x, y) = \frac{1}{2}(x^2 + y^2) + (1 - \mu)/r_1 + \mu/r_2 + \frac{1}{2}\mu(1 - \mu)$.

Decision space: N time slots, $q_i \in \{0, 1\}$, thrust direction is tangential, anti-tangential, or fixed. With K direction channels, $M = KN$ binary decisions.

Optimisation problem:

$$\min_{\mathbf{q} \in \{0,1\}^M} \left\| \mathbf{x}(t_f; \mathbf{q}) - \mathbf{x}_{\text{target}} \right\|_{\mathbf{w}}^2 + \alpha \|\mathbf{q}\|_1 + (\text{exclusion penalty})$$

where $\mathbf{x}(t_f; \mathbf{q})$ is the final state of the *nonlinear* CR3BP under schedule $\mathbf{q}$.

From STM linearisation to a QUBO

First-order STM expansion around a coast trajectory:

$$\mathbf{x}(t_f; \mathbf{q}) \approx \mathbf{x}_{\text{coast}}(t_f) + \sum_{j=1}^M q_j \mathbf{b}_j, \quad \mathbf{b}_j = \Phi(t_i, t_f) \mathbf{B} \mathbf{u}_{c,i} \Delta t.$$

Substituting into the squared-norm objective, with state gap $\mathbf{d} = \mathbf{x}_{\text{target}} - \mathbf{x}_{\text{coast}}(t_f)$:

$$E(\mathbf{q}) = \left\| \sum_j q_j \mathbf{b}_j - \mathbf{d} \right\|_{\mathbf{W}}^2 + \alpha \sum_j q_j = \mathbf{q}^\top \mathbf{Q} \mathbf{q} + \ell^\top \mathbf{q} + c_0$$

A QUBO with M logical qubits and dense couplings:

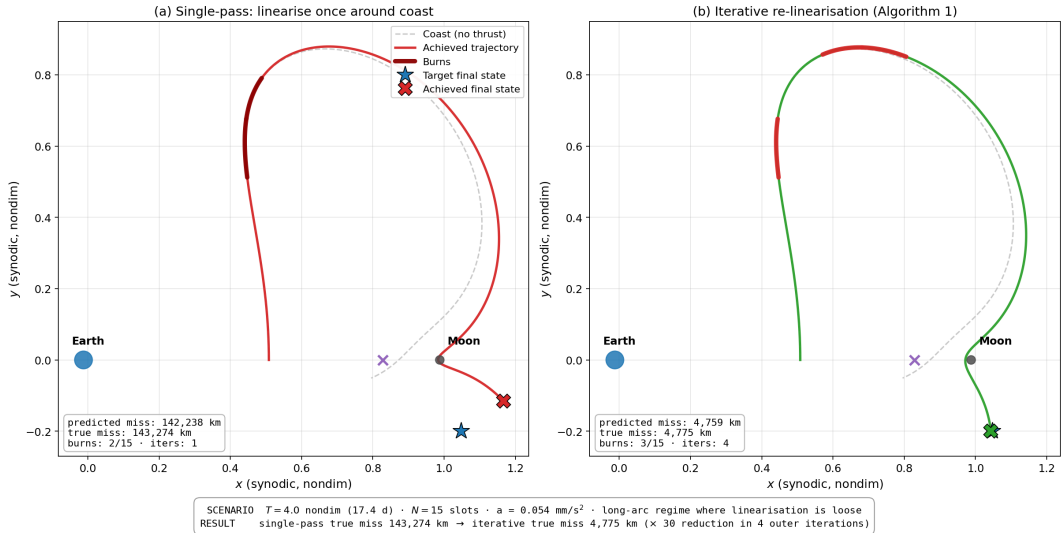
$$Q_{jk} = \mathbf{b}_j^\top \mathbf{W} \mathbf{b}_k$$

$$\ell_j = \alpha - 2 \mathbf{b}_j^\top \mathbf{W} \mathbf{d}$$

$$c_0 = \mathbf{d}^\top \mathbf{W} \mathbf{d} \text{ (energy offset)}$$

M scales *linearly* in the discretisation, vs. quadratically for pseudospectral-coefficient QUBOs.

The single-pass problem: linearisation gap



Algorithm 1: iterative re-linearisation with trust region

Initialise $\mathbf{q}^{(0)} = \mathbf{0}$ (coast); compute $r^{(0)} = \|\mathbf{x}(t_f; \mathbf{0}) - \mathbf{x}_{\text{target}}\|$.

Loop $k = 0, 1, \dots, k_{\max} - 1$:

Build QUBO linearised around $\mathbf{q}^{(k)}$.

Sample candidate $\hat{\mathbf{q}}$ (BF, MILP, SA, Kerberos, QPU).

If $\hat{\mathbf{q}} = \mathbf{q}^{(k)}$: terminate (fixed point).

Propagate nonlinear CR3BP under $\hat{\mathbf{q}}$; obtain $\hat{r}$.

Trust-region: if $\hat{r} > r^{(k)}$, terminate (return $\mathbf{q}^{(k)}$).

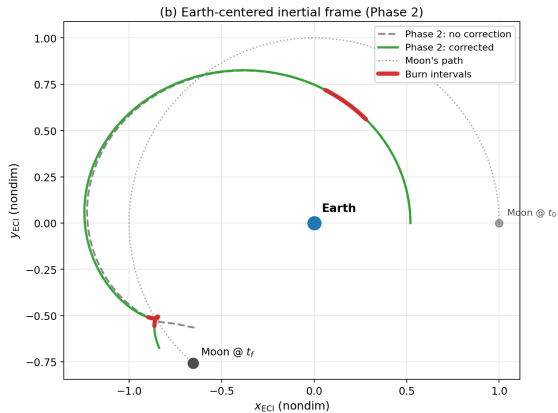
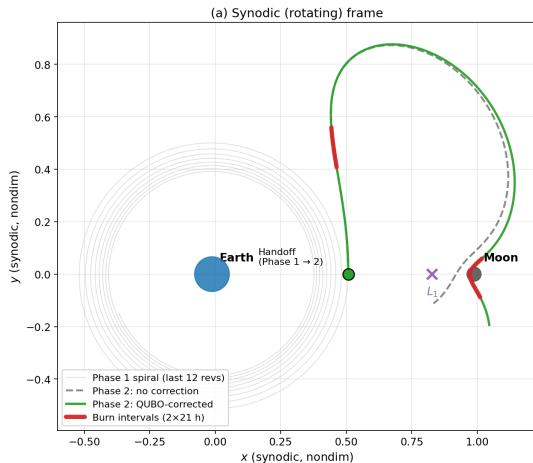
Set $\mathbf{q}^{(k+1)} = \hat{\mathbf{q}}$, $r^{(k+1)} = \hat{r}$.

If $r^{(k)} - r^{(k+1)} < \varepsilon_{\text{tol}}$: terminate.

Proposition 1 (monotonicity)

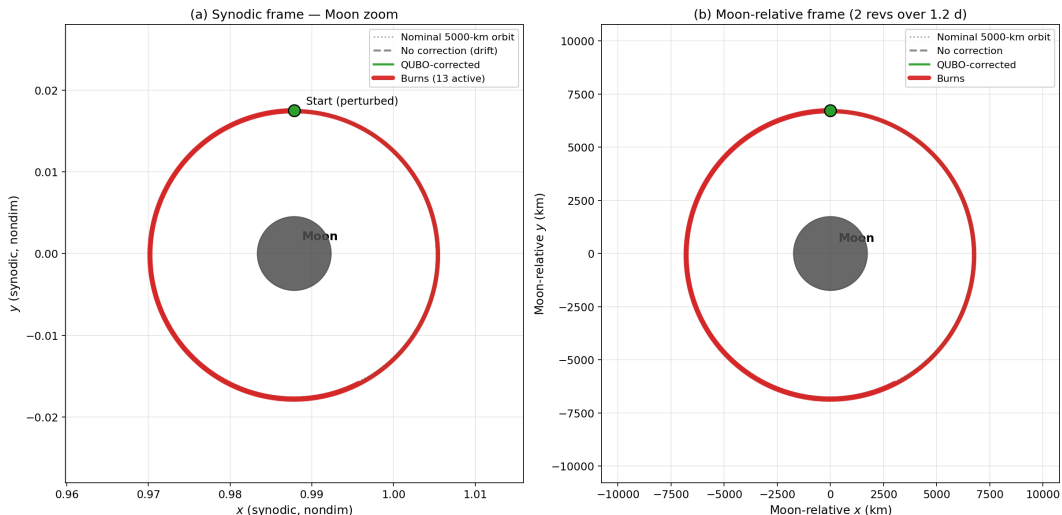
The sequence $r^{(k)}$ is non-increasing, $r^{(0)} \geq r^{(1)} \geq \dots \geq r^{(K)}$. The algorithm terminates in finitely many iterations, and the returned schedule is at least as good as coast: $r^{(K)} \leq r^{(0)}$.

Phase 2 results: $30\times$ improvement at 16 m s^{-1}



MISSION SPACECRAFT mass 100 kg · thrust 5.4 mN · $a = 0.054\text{ mm/s}^2$
 Phase 2 duration 4.0 nondim (17.4 d) · slot 21 h · $\Delta v_{\text{slot}} = 4.1\text{ m/s}$

Phase 4: lunar-orbit station-keeping



SPACECRAFT mass 100 kg · thrust 5.4 mN · $a = 0.054 \text{ mm/s}^2$ · single anti-tangential channel
 MISSION 5000-km lunar orbit · 0.5% tangential velocity perturbation at t_0 · horizon 0.265 nondim (1.2 d, 2 revs)

Phase 3: the multi-revolution challenge

Phase 2 leaves the spacecraft hyperbolic at perilune: $v_{\text{peri}}/v_{\text{circ}} \approx 1.49 > \sqrt{2}$.

To capture: shed $\sim 67 \text{ m s}^{-1}$ of excess velocity. With main thrust (3.5 mm s^{-2}) over a ~ 5 -hour close approach: $\Delta v_{\text{available}} \approx 63 \text{ m s}^{-1}$ — *just barely enough*.

Single-arc QUBO would fail: the linearisation gap over a multi-revolution capture exceeds the trust-region tolerance.

Solution: a *sliding-window* structure. Each window is a short arc with its own iterative re-linearisation; between windows the spacecraft drifts under true CR3BP.

Sliding-window: window cap + action-radius gate

Window cap: $\Delta T_{\max}$ fixed (e.g. 0.01 nondim ~ 1 h) so that the local linearisation remains faithful even when the orbit is loose ($T_{\text{loc}} = 2\pi\sqrt{r^3/\mu}$ diverges; we cap the window).

Action-radius gate: skip QUBO scheduling when the spacecraft is far from the Moon (Earth-perturbed regime). Drift freely instead.

Per-window target (velocity-only):

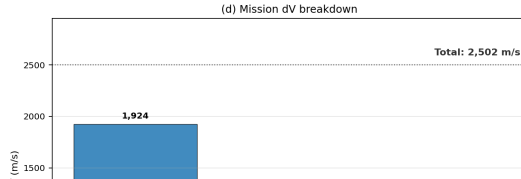
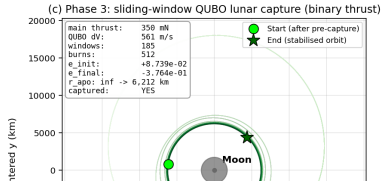
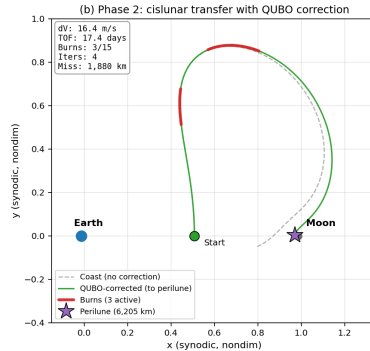
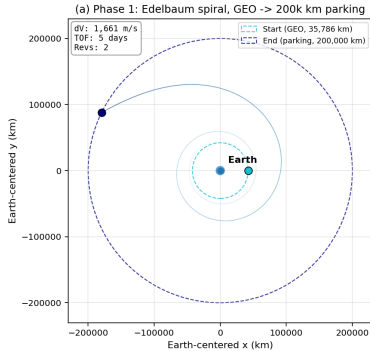
$$\mathbf{x}_{\text{target}}^{(w)} = [\mathbf{r}_{\text{eow}}, v_{\text{circ}}(\|\mathbf{r}_{\text{eow}} - \mathbf{r}_{\text{Moon}}\|) \hat{\mathbf{v}}_{\text{Moon-rel}}^{\text{eow}}]$$

“Stay where the unforced trajectory takes you, but slow to local circular speed.”

Effect: anti-tangential channel fires near perilune (where $v > v_{\text{circ}}$); stays off at apolune (can't speed up). Energy decreases monotonically per accepted window.

End-to-end mission overview

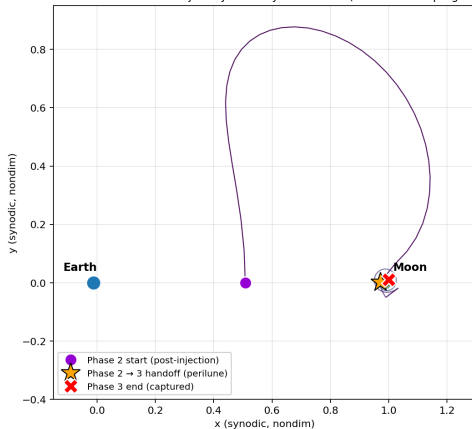
All-binary Earth-Moon mission: 100 kg microsatellite, throttleable ion/Hall thruster (350 mN main / 5.4 mN cruise)



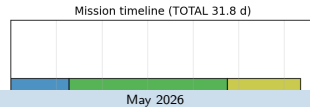
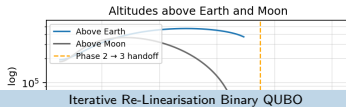
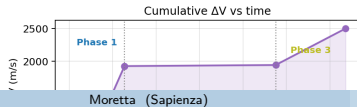
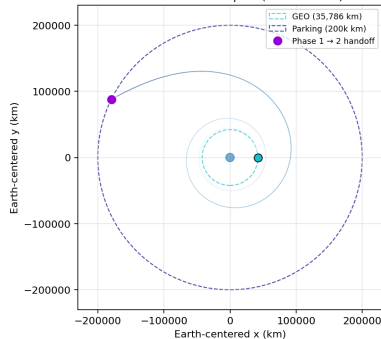
Continuity across phases

All-binary Earth-Moon mission: continuous trajectory across Phases 1 → 2 → 3

Phases 2 + 3: continuous trajectory in the synodic frame (colour = time progression)



Phase 1: Edelbaum spiral (Earth-inertial)



Mission Δv breakdown

Phase	Δv (m/s)	TOF (d)	Mode
1: Edelbaum spiral + injection boost	1924	6.4	main, tangential
2: Cislunar correction (QUBO)	16	17.4	cruise, tangential
3: Sliding-window QUBO capture	561	8.0	main, anti-tangential
Total	2502	31.8	all binary

Phase 1 dominates the budget (Earth gravity well).

Phase 3 second largest cost (capture).

Phase 2 is essentially free in Δv , but *enables the rest of the mission* — without correction the 22 000 km miss precludes any usable perilune passage.

All Δv delivered by binary on/off commands of one thruster.

Sliding-window parameter sensitivity

Configuration	Δv (m/s)	burns	$r_{\text{apo}}^{(f)}$ (km)
$\Delta T_{\text{max}} = 0.005$	2.6	300	14488
$\Delta T_{\text{max}} = 0.040$ (<i>baseline</i>)	11.9	175	13660
$\Delta T_{\text{max}} = 0.080$	33.8	255	12824
brake = 0.80	34.0	499	13194
brake = 1.00 (<i>baseline</i>)	11.9	175	13660
brake = 1.20	4.0	59	14128
action r : any value ≥ 0.04 identical (gate not triggered for bound orbit)			

Window cap is the elbow of the linearisation/throughput trade-off.

Brake factor trades fuel efficiency vs. apolune-shrinkage rate.

Action radius is a safety net: zero effect once the orbit is bound.

Comparison with state-of-the-art (NLP)

Benchmark vs. in-house Hermite-Simpson direct collocation ($T = 3$ nondim, $v_{\text{inj}}/v_{\text{circ}} = 1.187$):

Method	Miss (km)	Δv (m/s)	Solve time (s)
QUBO iter. ($N = 15$)	1 087	8.2	4.6
NLP HS-direct	0.0	8.8	3.2

NLP wins on accuracy: zero miss at comparable Δv .

NLP solution is a *continuous-thrust* schedule; requires post-processing to pulse-mode (re-introduces a discrete-action gap).

QUBO schedule is *directly executable* on a pulse-driven engine.

QUBO trade: action-space realism for tractability.

NLP is the right tool when thrust can be modulated continuously; QUBO is the right tool when the engine is genuinely binary.

Where this can compete

Niche fit (3–5 years horizon):

Cubesat / microsat with pulse-mode propulsion (no throttling) — direct QUBO match.

Real-time on-board correction (QUBO is pre-built; sampler can run in milliseconds).

Constellation / swarm scheduling (combinatorial decision space).

Hard discrete constraints (exclusion windows, time windows).

Where NLP / DDP / indirect methods will keep winning:

Globally Δv -optimal interplanetary missions (BepiColombo, JUICE).

Continuously throttled thruster optimisation.

Mature commercial frameworks (GPOPS-II, GMAT, Copernicus).

Differentiator: trust-region rigorous + naturally-binary + hardware-quantum-ready in one framework.
Each piece exists elsewhere; the combination is novel.

Limitations and future work

Within this paper:

Phase 3 reaches ~ 4500 km altitude bound orbit, not the 100 km target LLO. Continuing the sliding-window further reaches LLO at additional Δv .

Single-channel for Phases 2 and 3; multi-channel (tangential + normal + binormal) supported but not swept parametrically.

Planar CR3BP only; 3D extension is mechanical (no methodological change).

Natural follow-up:

D-Wave Leap Hybrid / Advantage-2 hardware results (sampler interface ready; only credentials missing).

Four-body model (Earth–Moon–Sun) admits weak-stability boundary trajectories with near-parabolic perilune speed.

Adaptive sliding-window (window length, brake factor, action radius chosen autonomously).

Conclusions

Naturally-binary QUBO formulation of low-thrust cislunar scheduling, complementary to Carbone et al.'s continuous-coefficient formulation.

Iterative re-linearisation with trust-region: *formal monotonicity guarantee* (Proposition 1) that single-pass solves do not have.

Sliding-window extension chains short arcs across many revolutions, with a window-length cap and an action-radius gate that confine scheduling to the linearisation-faithful regime.

One spacecraft, one throttleable thruster, four phases: end-to-end Earth–Moon mission delivered in ~ 32 d at $\Delta v_{\text{total}} \approx 2502 \text{ m s}^{-1}$, *all binary*, no impulsive burns.

Test suite: 264 / 264 passing. Paper: 20 pages, 5 figures, 5 tables. All code reproducible from `scripts/run_full_mission_demo.py`.

Thank you

Questions?

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